



DEALER BEST PRACTICE SERIES

Backlog Management -
Setting Effective Backlog
Execution Targets

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Backlog Management - Setting Effective Backlog
Execution Targets0
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1.0 Introduction

Backlog Management, as described in the Cat Global Mining M&R process is a very dynamic function. It requires constant attention and effort to manage and control the different phases of the process in order to meet the target of machine pending backlogs (risk) which is one of the major influencers on % of Scheduled Downtime.

The primary objective of the detection routines (Condition Monitoring plan) is the timely detection of any abnormal condition that can develop into an unscheduled event or failure, and for Backlog Management is the efficient and effective transformation of the input / alerts from Condition Monitoring into effective repair / corrective actions to be scheduled before failure.

At this time, it is convenient to remember our definition of Backlog (BL). Any defects that have been detected and the repair can be deferred is a Backlog ... if we have a defect entered in our records, then we are already late and need to *Act*. Backlog Management is the efficient and effective management of the total “*To Do*” List.

This definition supports the dynamic of the Backlog Management process, making the organization act promptly in a presence of a Backlog *Analyzing – Planning – Allocating Resources – Scheduling the Execution and finally Executing it* to fix the abnormal condition detected to eliminate the risk of a break down.

The number of pending backlogs in a fleet of equipment or in a particular machine is the result of the balance maintained between Backlogs Generated – Backlogs Pending – Backlogs Executed. The close control of these three parameters is the key to managing the backlog workload, simple to do it in a small fleet of machines but it can be very difficult when the fleet size gets incremented and / or we lose control of the Backlog workload balance.

The best practice herein presented describes the preferred methodology / procedure to set up effective backlog execution targets to guide the organization in meeting the expected pending backlog goal.

For better understanding and integration with the rest of the M&R processes consider working this Best Practices in conjunction with the BP 1110-1.03-1261, “Unavailability Distribution”.

2.0 Best Practice Description

As mentioned before, the main objective is to maintain the balance between BL Generated – Executed – Pending to achieve the end goal of minimum unscheduled events. In the understanding that unscheduled events or failures can be minimized by keeping the risk of failure low through a proactive Detection & Backlog Management strategy, this Best Practice provides the M&R recommended methodology to keep the maintenance organization focused through determining and observing effective Backlog execution targets.

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1.- Determine Total Backlog Pending Goal

As we already mentioned, pending backlogs is equal to risk of failures or unscheduled events. So it is very obvious that we need to keep this number as low as possible. But remember at the time of setting a goal or target that we need to analyze and consider the current status of the equipment condition / Backlog Pending.

Define:

Target for Pending Backlogs / Equipment = A
 Total Pending Backlogs = A x Number of Machines

Considerations:

For New Machines

Participate in the assembly of the machines and ensure the machines are delivered with zero ("0") Backlogs. Bring the machine back to the shop within the first 200 hours of operation and clean all possible backlogs / defects that may have been developed as the results of the first operating hours.

For new machines, the recommended Pending BL Goal should be "0" (zero). This in fact will guide your organization to keep an eye on the machines and not only rely on the performance of new equipment to support the availability.

Pending Backlog / Equipment = 0
 Total Pending Backlogs = 0 x Number of Machines = 0

Yes it is a very aggressive goal, but goals are made for "moving", to promote *actions of the* organizations in the right direction.

For Mature Machines

In the case of mature machines, it is very challenging to maintain the above goal of "zero Backlogs", and if taken the pursuit of it must be supported by very effective execution of the detection routines, efficient Backlog Management system and a very well controlled quality of all repairs that are being executed. At any time, the machine gets worked on by the maintenance crew it must be returned to operation with zero backlogs pending.

Effective operations have proven that it is possible to maintain the pending backlogs in control at levels below 5 BL. We recommend setting your targets in the range of 0 to 5 BL pending per machine.

When Pending Backlog is out of control

In this case, when Pending Backlogs is out of control and the machines have several accumulated defects, the recommended strategy is to use an escalating targets strategy. This will

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help in avoiding the frustration of the organization that may be produced as a result of not being able to achieve the desired goal in the expected timeframe.

It is also convenient in this type of situations to conduct a thorough inspection of your fleet to ensure the information available by planning & scheduling is as accurate as possible. Remember that in this type of situation, it is normal for the detection level gets negatively affected.

Note: To develop the description of the process of setting the BL execution targets, we will use the same case scenario / information used in the “Unavailability Distribution” Best Practice, BP1110-1.03-1261.

Equipment Type	793 OHT
Number of Equipment	30
Physical Availability Goal	87%
Utilization of Asset	81%
Month	January

With a final adjusted Unavailability Distribution of (see BP1110-1.03-1261):

Unavailability Distribution	Percentage	Hours
Preventive Maintenance	3.24%	723
Planned Component Replacement (PCR)	1.21%	270
Major / Minor Scheduled Repairs	0.43%	96
Detection Routines	0.33%	74
Tires	1.33%	297
Other Repair Centers	0.11%	24
Unscheduled Repairs	6.92%	1545
TOTAL UNAVAILABILITY	13.57%	3029
ESTIMATED AVAILABILITY	86.43%	19291
TOTAL HOURS TO EXECUTE M&R EVENTS		2902
REMAINING HOURS FOR M&R EVENTS		-128
AVAILABILITY GOAL	87.00%	19418

From BP1110-1.03-1261:

“The above chart is the representation where we applied some of the adjustment ideas. We eliminated 1 PCR, reduced the M&M time by reducing the allocated time for M&M by frequency

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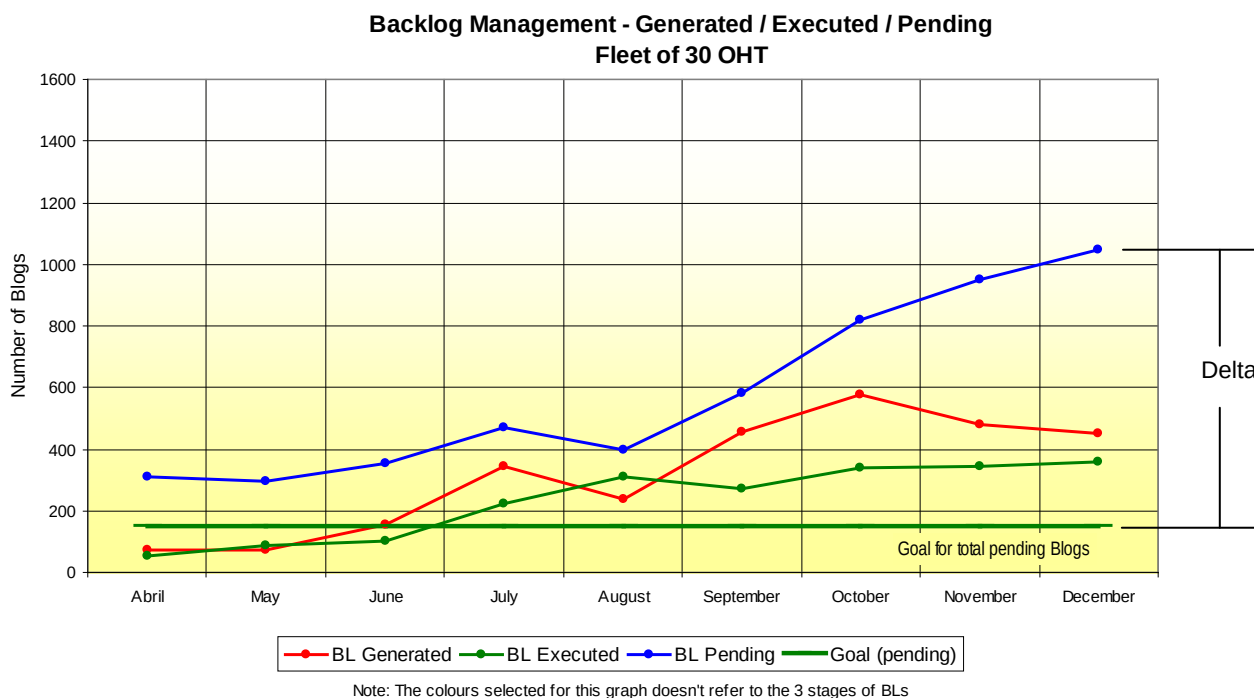
with the agreement to execute more backlogs in parallel with the PM services, and increase the target for using the windows of opportunities for inspections. Reduction of the estimated time for the unscheduled repairs is not suggested; it would be just a mathematical adjustment. What we can do with the unscheduled repairs is to ensure that the unscheduled hours don't increase and to set effective machine performance targets for the scheduled repairs to be performed. This will give better control of the future unscheduled repairs reduction."

2.- Analyze the balance between BL Generated – Executed – Pending

The Backlog Management information we will use for this case scenario is the following:

Truck fleet - Backlog Management Information										
	Abril 30	May 31	June 30	July 31	August 31	September 30	October 31	November 30	December 31	January 31
Number of Trucks	30	30	30	30	30	30	30	30	30	30
Hours in the Month	21600	22320	21600	22320	22320	21600	22320	21600	22320	22320
Backlogs/ Machine	10	10	12	16	13	19	27	32	35	
Backlogs Generated	75	71	157	345	236	457	576	478	450	
Backlogs Executed	55	87	101	225	312	270	340	345	357	
Backlogs Pending	312	296	352	472	396	583	819	952	1045	
Goal Pending BL/Machines	5	5	5	5	5	5	5	5	5	
Goal for Total Pending BL	150	150	150	150	150	150	150	150	150	

A picture of this data revealed more clearly the situation:



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The graphical representation of the data very quickly highlights the challenge that we are facing. In the last 4 months, the BL Generation has been greater than the BL Execution, ending the month of December with a negative delta of 895 BL for the whole fleet, or an excess of 29.83 BL per machine.

The deviation to be corrected: Current Pending BL + BL Generation (Est.) – Pending BL Goal
 = 1045+ 464 (average of last two months) – 150
 = 1359

So, the first step is to recognize we have a problem, the second step will be ...”how to solve it”. There are many questions that can be raised at this time, such as:

- How did we end up in this situation?
- Why we didn't react sooner?
- Is this situation already affecting our % scheduled work (or % Break Downs)?
- Why were we not able to increase the execution rate?
- Do we have parts problems?
- Do we have resources problems?
- ...

All valid questions, but now is the time to fix this situation. How can we guide our organization to focus their efforts to correct this situation? The answer is ... **Effective BL Execution Targets**.

3.- Determine the BL Execution Targets

Backlog Execution Targets will have to address 2 issues, How Many Backlogs we will have to execute and Who will do It.

Let's start with the most frequent activity in our maintenance operation, PM services.

3.1 Backlogs to execute in PM services

First determine the PM Service time for the period, normally one month. The Unavailability Distribution will provide this information.

Unavailability Distribution		Percentage	Hours
Preventive Maintenance		3.24%	723

We have a projection of 723 hours for all our PMs for the month of January. Now, how many Backlogs we can possibly execute in parallel without affecting the efficiency of the PM?

The answer is in the PM Backlog Execution Window and in the Backlog Average ETTR (Estimated Time To Repair).

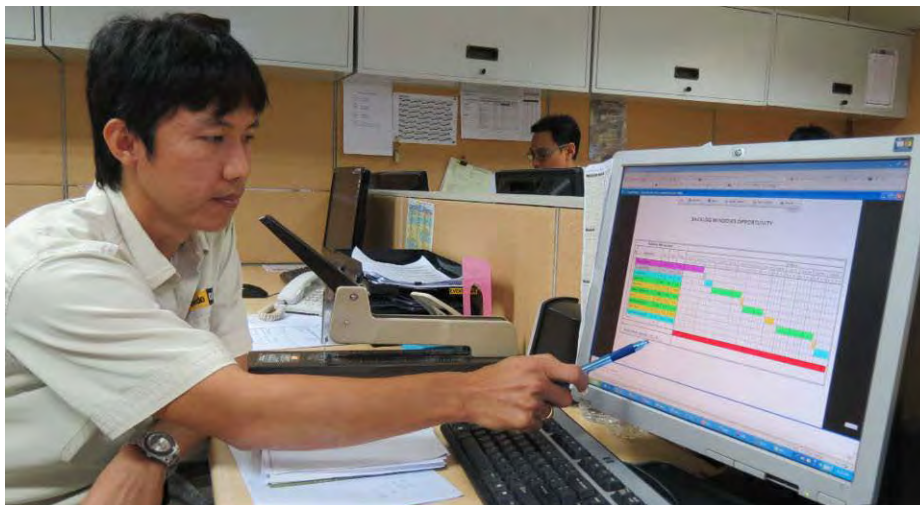
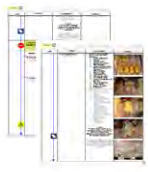
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PM Backlog Execution Window

The BL execution window within the PM services needs to be determined by analyzing the sequence of activities to be executed and specifically determining the time the machine is de-energized (engine shut off). The use of the 8 Step PM will be of great advantage; in this case the BL execution window is always constant.

In the case study example we are using an 8 step 10 hours constant duration for all PM services. The analysis of the activities to be performed during PM will provide for your operation the final scheduling window for Backlogs. The key is to determine the time interval in where the machine being serviced is de-energized.

For our example we will use 6 hours (60% of the total PM Service duration) as our scheduling windows.



In this illustration the on site planner at Batu Hijau – Trakindo is working in the definition of the Backlog Scheduling Window.

Now we have to determine the average estimated time to repair (ETTR) of the backlogs in our records. The ETTR records are as follows:

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	Abril	May	June	July	August	September	October	November	December
BL ETTR	218.4	207.2	246.4	330.4	277.2	408.1	573.3	666.4	731.5
ELTR	436.8	414.4	492.8	660.8	554.4	816.2	1146.6	1332.8	1463
Total available Labor H	1500	1500	1500	1500	1500	1500	1500	1500	1500
10 % of T.avail labor	150	150	150	150	150	150	150	150	150

ETTR = Estimated Time To Repair, ELTR = Estimated Labor To Repair

Based on the month of December information the average ETTR (Estimated Time To Repair) for the pending Backlogs is:

BL Total ETTR	732
Total Pending Backlogs	1045
BL average ETTR	0.70 h

With this information we can determine the BL execution target for the PM services for the month of January.

Total PM hours month	723 h
Month BL execution window	434 h
BL Total ETTR	732
Total Pending Backlogs	1045
BL average ETTR	0.70 h
BL Execution Target - PM	620 BLs
BL to execute / PM	8.6

The 620 BLs to execute is the result of dividing the BL execution window by the BL average ETTR. BL exec. Target – PM = 434 h / 0.70 h = 620 BLs

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This is our first target.

Unavailability Distribution	Scheduling Windows		Hours to Execute Backlogs	BL Execution Target
	Percentage	Hours		
Preventive Maintenance	3.24%	723	434	620
Planned Component Replacement (PCR)	1.21%	270		
Major / Minor Scheduled Repairs	0.43%	96		
Detection Routines	0.33%	74		
Tires	1.33%	297		
Other Repair Centers	0.11%	24		
Unscheduled Repairs	6.92%	1545		
TOTAL UNAVAILABILITY	13.57%	3029	Total Backlogs To Execute	620

PCR Backlog Execution Window

In the case of the PCR's the Backlogs that we can execute will be determined by the current backlogs per machine times the number of PCR's that we will perform in the period. The goal is to execute 100% of the Backlogs pending on the PCR machines.

In this case after the adjustments done to the projected unavailability distribution we ended with 2 PCR's with a total duration of 270 hours or 11.25 days (See page of BP BP1110-1.03-1261).

The Backlogs that can be executed in this window are: 2 (PCR's) x 35 (BLs per machine)
= 70 BLs

Unavailability Distribution	Scheduling Windows		Hours to Execute Backlogs	BL Execution Target
	Percentage	Hours		
Preventive Maintenance	3.24%	723	434	620
Planned Component Replacement (PCR)	1.21%	270	---	70
Major / Minor Scheduled Repairs	0.43%	96		
Detection Routines	0.33%	74		
Tires	1.33%	297		
Other Repair Centers	0.11%	24		
Unscheduled Repairs	6.92%	1545		
TOTAL UNAVAILABILITY	13.57%	3029	Total Backlogs To Execute	690

With the addition of Backlogs to execute during PCR's the total number of Backlogs that we are projecting to execute is 690. But the correction needed is 1359 (page 5), still not enough.

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Detection Routines

In this case we will not try to force a specific number of backlogs to execute, but we will recommend our inspectors to contribute with the correction of small defects that does not affect the performance or their primary role of defect detection. In this scenario in where the % of unscheduled DT is significant we need to be careful not to deviate the focus / attention from the main objective of critical functions such as the detection routines.

We will not assign any target of BL execution to this routine. We recommend you to analyze your particular situation and take the best suitable decision for it.

Tires & Other Repair Centers

Normally these types of jobs are being taken by another crew / workshops. Because of that and because of safety reasons we don't recommend performing work in parallel. Normally that's the case, exceptions need to be evaluated at your site and the decision needs to be taken at that time.

We will not assign any target of BL execution to these routines. The BL target situation at this stage stays the same.

Unavailability Distribution	Scheduling Windows		Hours to Execute Backlogs	BL Execution Target
	Percentage	Hours		
Preventive Maintenance	3.24%	723	434	620
Planned Component Replacement (PCR)	1.21%	270	---	70
Major / Minor Scheduled Repairs	0.43%	96		
Detection Routines	0.33%	74	0	0
Tires	1.33%	297	0	0
Other Repair Centers	0.11%	24	0	0
Unscheduled Repairs	6.92%	1545		
TOTAL UNAVAILABILITY	13.57%	3029	Total Backlogs To Execute	690

Unscheduled Repairs

In the case of Unscheduled Repairs we need to analyze the distribution of the unscheduled events and determine the amount of work that is being executed in the field and in the workshop. The Field Service normally takes care of the small repairs in the field. Repairs that don't require opening components, compartments, systems because of Contamination Control issues and also those that is easy and quick to solve. If a component or system needs to be open and/or the estimated time to repair recommend not to execute in the field are send to the workshop.

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We will not assign any tasks for the work to be done in the field but we will do so for the unscheduled repairs that are executed in the workshop.

The case scenario that we are going to use presents the following distribution for the unscheduled repairs (determine your particular situation in your operation):

Field MTTR	20.0 min
Stops in the Field	907
Total Hours for the Month	302
Shop MTTR	5.7 h
Stops in the Shop	218
Total Hours for the Month	1243
Total Unscheduled Down Time	1545

Note: Remember to define and use your specific operation situation.

In order to determine the quantity of Backlogs to execute in this unscheduled repairs window we need to consider the following issues:

- The work in parallel cannot represent a disruption for the execution of the main repair that in this case is the unscheduled or break down repair.
- Availability of labor.
- Backlogs in "Green" or Ready to Go.

The recommendation is to analyze every unscheduled stop together with Planning & Scheduling to determine if any work can be done in parallel and also to define the specific backlog to execute.

We will take a very conservative decision and will start with a target of Backlog execution of 5% of the total Shop - Unscheduled DT window.

Total Shop Hours for Unscheduled Repairs	1243
% of DT to use for Backlog Execution	5.0%
Hours to Target for BL Execution	62.15

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And the number of Backlogs to execute in this window will be:

Backlogs Average ETTR (h)

0.7

Backlogs to Execute

89

At this time the scenario for the Backlog Execution targets looks like:

Unavailability Distribution	Scheduling Windows		Hours to Execute Backlogs	BL Execution Target
	Percentage	Hours		
Preventive Maintenance	3.24%	723	434	620
Planned Component Replacement (PCR)	1.21%	270	---	70
Major / Minor Scheduled Repairs	0.43%	96		
Detection Routines	0.33%	74	0	0
Tires	1.33%	297	0	0
Other Repair Centers	0.11%	24	0	0
Unscheduled Repairs	6.92%	1545	62	89
TOTAL UNAVAILABILITY	13.57%	3029	Total Backlogs To Execute	779

We have a target of 779 Backlogs to execute, but as stated in page 5 the correction needed is 1359. It is time to open another window to fit in the required delta of 580 BL's.

M&M Backlog Execution Window

The primary use of the Major & Minor Repairs window is to balance the Backlog workload executing those Backlogs that cannot be fitted into the other scheduling windows. The two primary windows for Backlog execution are PM services and PCR's. The third window to consider is unscheduled work that because of its nature it is very difficult to get full advantage of them. So, when we face a situation in where the PM and PCR's windows are max out in terms of Backlog execution, the strategy is to open another window tailored to fit the additional Backlogs to execute. This window is the Major & Minor Repairs window.

The Planner has to design the length and frequency of the M&M window having four variables to play with, the amount of Backlogs to execute, the % of unavailability, the work force (labor) required and the timeframe to fix the problem and get the Backlog situation in control.

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In this case the M&M window required to fit the required 580 Backlogs is:

M&M Repair Window	406.3 h
M&M % of unavailability	1.82%

A 1.82% of unavailability would put us well below the availability target (87%; see BP 1110-1.03-1261) we have to comply with.

Unavailability Distribution	Scheduling Windows		Hours to Execute Backlogs	BL Execution Target
	Percentage	Hours		
Preventive Maintenance	3.24%	723	434	620
Planned Component Replacement (PCR)	1.21%	270	---	70
Major / Minor Scheduled Repairs	1.82%	406		
Detection Routines	0.33%	74	0	0
Tires	1.33%	297	0	0
Other Repair Centers	0.11%	24	0	0
Unscheduled Repairs	6.92%	1545	62	89
TOTAL UNAVAILABILITY	14.96%	3339.3	Total Backlogs To Execute	779
ESTIMATED AVAILABILITY	85.04%			

At this stage we need to make a decision based on three options:

- Not open another window in favor of the availability target
- Open a window of 406 hours and take an extra 1.82% of availability pushing the estimated availability target down to 85.04 %
- Reduce the M&M repair window to a percentage that doesn't affect too much the availability and can be negotiated with the customer or production department.

Now it is starting to be clear that the elephant (Backlog pending) cannot be eaten in one month.

It is a challenge, but if we want to improve the unscheduled DT situation (6.92%) we need to reduce the risk of failures by cleaning the pending Backlog list. So in this example we will go with the alternative 3 of using a reduced window (less hours / percentage) for the M&M repairs.

To complete the example we will use the adjustment done by the planner in the example presented in the BP BP1110-1.03-1261, in where the final percentage of unavailability for the M&M repair window was determined to be 0.43%.

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When the M&M window was adjusted basically we reduced the duration of this window from 4 to 2 hours, this was done in the final adjustments introduced to try to meet the availability target.

By Frequency				By Frequency			
M&M Sched. Repairs Frequency	500	h		M&M Sched. Repairs Frequency	500	h	
M&M Sched. Repairs Duration	4	h		M&M Sched. Repairs Duration	2	h	
By Exception				By Exception			
Major Repairs qty	2			Major Repairs qty	2		
Duration	12	h/each		Duration	12	h/each	

When you end up with a situation like this in where the M&M window / duration is very limited (2 hours) the best strategy is to add this window to the current PM services. It is convenient to mention that when the M&M window is larger it is beneficial to open a separate window normally in between the PM services. This will increase the frequency that the service crew will be able to “see” and “touch” the machine and have more chances to prevent unexpected failures (Detect – Plan – Execute).

The time duration and the number of Backlogs to execute in this window will be:

Total M&M repairs by frequency	36.2	
Duration	72.3	h
Backlogs to Execute in this window	103	
Repair Duration by Exception	24.0	h
Repairs (Backlogs) to Execute	2	
Total Duration	96.3	h
Unavailability due to M&M repairs	0.43%	
Total Backlogs to Execute in the M&M Window	105	

In the case of the specific repairs of 12 hours (2 Machines for 12 hours each) the decision to make is whether to attach these repairs to a PM Service or to perform a complete different stop. The urgency of the repairs will tell. When deciding to execute the repairs in a different stop the time for washing needs to be considered.

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The unavailability distribution and the Backlog Targets look like this:

Unavailability Distribution	Scheduling Windows		Hours to Execute Backlogs	BL Execution Target
	Percentage	Hours		
Preventive Maintenance	3.24%	723	434	620
Planned Component Replacement (PCR)	1.21%	270	---	70
Major / Minor Scheduled Repairs	0.43%	96	96	105
Detection Routines	0.33%	74	0	0
Tires	1.33%	297	0	0
Other Repair Centers	0.11%	24	0	0
Unscheduled Repairs	6.92%	1545	62	89
TOTAL UNAVAILABILITY	13.57%	3029.316	Total Backlogs To Execute	884
ESTIMATED AVAILABILITY	86.43%			



With this target of 884 Backlogs to execute we are at 65% of the total correction needed (1359 Backlogs). As we mentioned before the total correction cannot be made in one single month.

To determine the final timeframe (months) that we will use to complete the corrections needed and bring the situation of Backlogs pending back in control, we have to analyze the resources needed and available in terms of labor, parts, facilities and equipment.

Important to notice is that even though we cannot correct the deviation in one month, the targeted backlog execution represent an increase of more than 200% compared with the current backlog execution (last month = 357 BL executed).

As mentioned before in the analysis of the Backlog Balance (Generated – Executed – Pending), there are several questions to investigate:

- How did we end up in this situation?
- Why we didn't react sooner?
- Is this situation already affecting our % scheduled work (or % Break Downs)?
- Why were we not able to increase the execution rate?
- Do we have parts problems?
- Do we have resources problems?

One thing is perfectly clear; we need to do something different, maintaining an execution rate of Backlogs below the amount that is being generated in the same period (357 executed vs. 450) will only make the situation worst. We have to stop digging and start filling the hole we created.

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Based on the data provided the workforce situation is a big challenge, the amount of estimated labor in the pending backlog is increasing rapidly and could pass the available labor if no corrections are introduced.

	Abril	May	June	July	August	September	October	November	December
BL ETTR	218.4	207.2	246.4	330.4	277.2	408.1	573.3	666.4	731.5
ELTR	436.8	414.4	492.8	660.8	554.4	816.2	1146.6	1332.8	1463
Total available Labor H	1500	1500	1500	1500	1500	1500	1500	1500	1500
10 % of T.avail labor	150	150	150	150	150	150	150	150	150

Most probably for this case the organization will have to make a decision to increase the overtime or request external support (labor force peak shaving) to overcome the situation.

We recommend at this stage to present the organization with the analysis of the situation and the plan to increase the Backlog execution following specific targets and accountabilities. Discuss the alternatives and implement.

Follow up closely the results, specially the achievement of the BL execution targets. Make the adjustments necessary, apply the same methodology for the following month and monitor the results to ensure the achievement of the target for total Backlog pending, achieve the balance between the BL Executed – Generated & Pending.

As mentioned before we need to do something different to turnaround the Backlog Pending situation otherwise we will be carrying a constant risk of having unexpected events (unscheduled or break downs). We have to stop digging and start filling the hole we are into.

3.0 Implementation Steps

The implementation of this concept is relatively simple. This is a concept and procedure that has to be the responsibility of the on site planner. It is a strategic job.

The steps for implementation are as follows:

- ✓ Evaluate your planning process. Determine the AS IS situation with respect to determine and use of the Unavailability Distribution and the Backlog Execution Targets.
- ✓ If not in use, adopt the Unavailability Distribution procedure described in the BP1110-1.03-1261.
- ✓ Determine the equipment fleet operating parameters (utilization, availability target ...).
- ✓ Investigate and gather all needed information.
- ✓ Populate your Unavailability Distribution and define the windows of opportunities to do work in parallel.
- ✓ Determine and analyze the Backlog Management situation.
- ✓ Apply the methodology here presented to determine the BL execution targets.

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- ✓ Present the concept and targets to your organization. This step is very important to get the “buy in” of your organization. The success of this procedure depends on the collective work of the on site organization.
- ✓ Incorporate the BL Execution Targets in all your planning meetings. Remember that the task of determining the BL Execution Targets is a task that should be (must) done by the planner, in some dealers called “Strategic Planning”.
- ✓ Link the Long, medium and short-term plans to the Backlog Management situation.
- ✓ Follow up. Closely follow up the execution and accomplishment of the targets defined.
- ✓ Always analyze the results of your Backlog Management together with the % Unscheduled Work or Breakdown DT and the Condition Monitoring KPI’s results specially “Detection Level”. A low detection level can easily ruin your Backlog Management balancing efforts.
- ✓ Manage as necessary.

4.0 Benefits

The definition and use of effective Backlog Execution Targets will provide a way to help focus the organization toward failure prevention. Specific targets provide a quantifiable way of assigning responsibilities and manage by results achievement.

For Planning:

- ✓ Clear goals and targets for following up and controlling the execution of the scheduled tasks.
- ✓ Better management of the adjustments that may be necessary to introduce.
- ✓ Gain control.
- ✓ Moving from expecting “good” things to happen ... to controlling and making it happen.

For the Work Shop crews (PM services, unscheduled, PCR’s, etc.)

- ✓ Clear set of goals to guide and meet the required backlog execution.
- ✓ Better information to make better decision (When and when not to extend or do extra repairs).
- ✓ Clear identification of opportunities.
- ✓ Better management of maintenance activities.
- ✓ Integration of the efforts with the rest of the areas of the maintenance organization.

For the Operation Department”

- ✓ Better response from the maintenance department to the operator’s feedback.
- ✓ Better, more reliable equipment to operate (Backlogs in control).

For the Production Department

- ✓ Backlogs in control → better, more reliable machines → better planning of production.
- ✓ Better machines.

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5.0 Resources Required

The implementation of this BP will not require any extra manpower or infrastructure, but depending on your particular situation it will be necessary to adequate or re-design the function and role & responsibilities of the planner. Special attention must be dedicated to the cases in where the planner gets primarily focused in building and structuring the short-term. Building the short-term plan is a scheduling function. If that's the case, we recommend separating the functions of planning and scheduling and assigning resources for both functions.

The procedure here presented in this BP is a responsibility of the planner. It is part of building strategic guidelines for the organization to follow.

6.0 Supporting Attachments / References

No attachments.

7.0 Related Best Practices

- **BP1110-1.03-1261** – “Unavailability Distribution”- Effectively Manage the Unavailability of your Equipment Fleet.
- **BP 1007-2.0-1103** – Backlog Management

8.0 Acknowledgements

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